

Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

- The purpose of this lab is to learn how deleted or damaged files can be recovered from a forensic storage image.
- It focuses on examining raw binary data and identifying file signatures.
- The lab also demonstrates how embedded and deleted files can be extracted.

2. Lab Report Structure

The report explains the steps used to examine, recover, and verify digital evidence. It includes the use of XXD for hexadecimal and file-signature analysis, It covers The Sleuth Kit for examining inodes, blocks, sectors, and partitions. It also documents Binwalk and Scalpel for embedded-file discovery and file recovery.

2.1. Title Page

- Lab Title: Lab 3 — Data Carving with XXD, Binwalk and Scalpel
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- Student ID: 2025/FWSD/11320.
- Course Name: SBT-DF202 — Computer and Digital Forensics
- Instructor Name: Aminu Idris
- Date of Submission: 4/09/2026
- Version:

2.2. Executive Summary

- This lab focused on analyzing forensic images and recovering files that may not be visible through normal browsing. XXD was used to examine hexadecimal data and identify JPEG headers and footers. The Sleuth Kit and Binwalk were used to examine storage data and identify embedded content.

Scalpel was used to recover deleted files from the USB forensic image and recovered evidence was verified using hashes and documented for forensic reporting

2.3. Lab Objectives

- The lab aims to explain why data carving is necessary when file-system information is missing or unreliable.
- It teaches students to identify file signatures and examine binary data using XXD.
- It develops skills in using The Sleuth Kit, Binwalk, and Scalpel for forensic analysis.
- Students also learn to recover deleted files and verify evidence using cryptographic hashes.
- The final objective is to produce a clear and professional forensic report.

2.4. Tools and Resources Used

- The main tools used are **XXD, Binwalk, Scalpel, and The Sleuth Kit**.
- The Sleuth Kit includes tools such as `img_stat`, `mmls`, `fsstat`, `fls`, `icat`, and `blkcat`.
- The lab uses **Ch01InChap01.dd** as a forensic DD image.
- We also use **J_ub_law.jpg** for hexadecimal analysis and **120M.7z** for file carving.
- MD5 and SHA-256 are used to verify the integrity of the evidence and recovered files

2.5. Methodology

Laboratory guidance

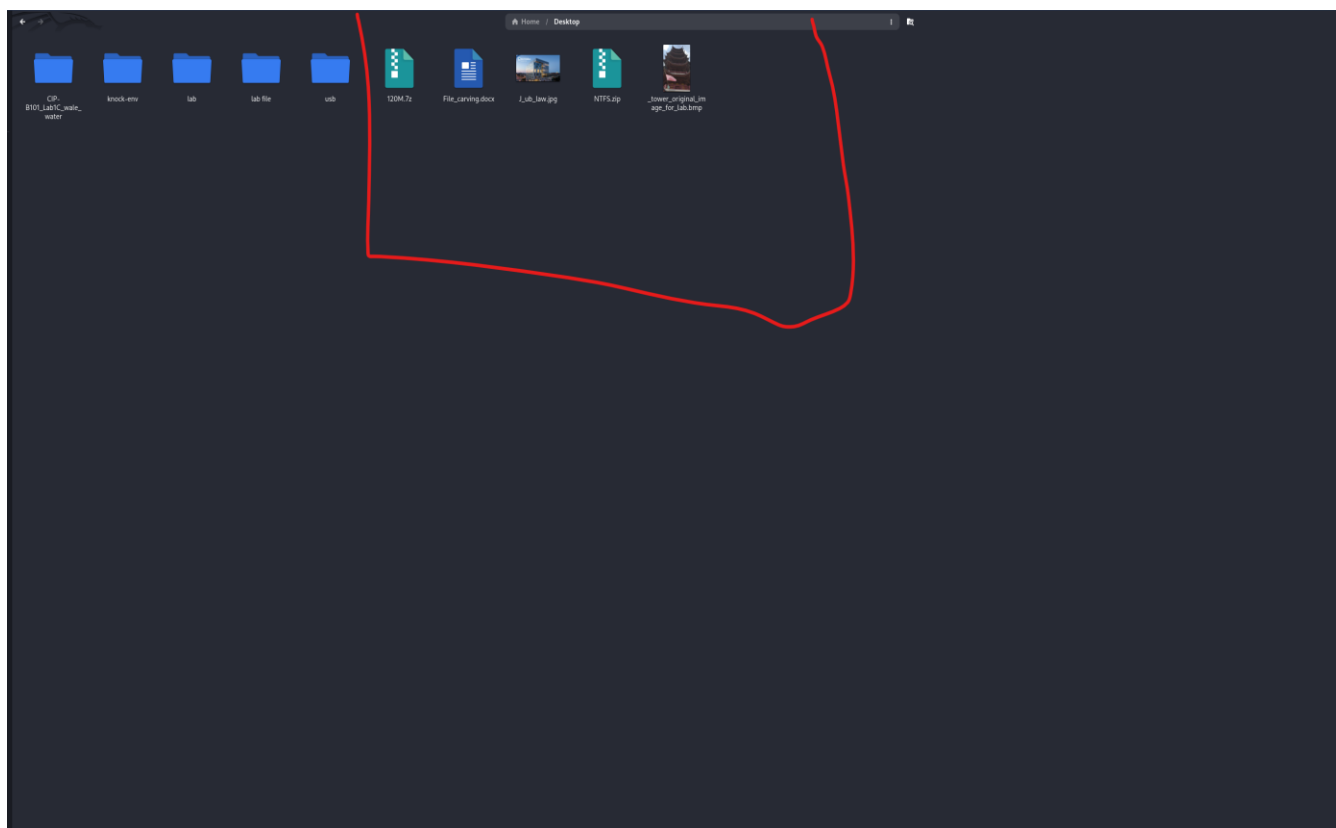
Before beginning this lab, review the **Module 7 lessons and study material on Data Carving and File Recovery Techniques**. The lecture material is already a detailed walkthrough, so you are expected to understand file signatures, hexadecimal analysis, embedded-file recovery and carving concepts before executing commands.

Create a dedicated working directory and preserve all downloaded evidence files. Record MD5 and SHA-256 hashes before beginning analysis.

Required Lab Resources

Download the authorised training evidence below:

- [Ch01InChap01.dd — Forensic DD Image](#)
- **J_ub_law.jpg — JPEG Sample for Hex and Signature Analysis**
- **120M.7z — USB Image for Scalpel File Carving**



```
water@kali:~/Desktop
$ ls
120M.7z  CIP-B101_Lab1C_wale_water  File_carving.docx  J_ub_law.jpg  knock-env  lab  'lab file'  NTFS.zip  _tower_original_image_for_lab.bmp  usb

water@kali:~/Desktop
$ file J_ub_law.jpg
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
```

Tools Installed for the assignment

Part A - File Signatures and Hex Analysis

Use xxd to inspect J_ub_law.jpg, identify the JPEG header and footer signatures, create a plain hexadecimal dump, and reconstruct the image using reverse-hexdump functionality.

Verify the reconstructed image by comparing MD5 and SHA-256 hashes with the original.

Create a plain hexadecimal dump, and reconstruct the image using reverse-hexdump functionality. Verify the reconstructed image by comparing MD5 and SHA-256 hashes with the original

Verifying the Hash value of the original image **83a360ac7f7e0ca318e5bfe39f95f137**

J_ub_law.jpg

```
(water@kali)-[~/Desktop]
$ md5sum J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
```

```
(water@kali)-[~/Desktop]
$
```

Creating the hexadecimal dump and Reconstructing the JPEG from the hex dump

```
(water@kali)-[~/Desktop]
$ xxd -p J_ub_law.jpg > image_hex.txt
```

```
(water@kali)-[~/Desktop]
$ xxd -r -p image_hex.txt reconstructed.jpg
```

```
(water@kali)-[~/Desktop]
$ ls -lh reconstructed.jpg
file reconstructed.jpg
```

```
-rw-rw-r-- 1 water water 1.7M Sep  4 05:45 reconstructed.jpg
reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
```

Verify the reconstructed image by comparing MD5 and SHA-256 hashes with the original.

The reconstructed hash value is

238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg

238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 reconstructed.jpg

```
(water@kali)-[~/Desktop]
$ md5sum J_ub_law.jpg reconstructed.jpg
sha256sum J_ub_law.jpg reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137  reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  reconstructed.jpg
```

```
(water@kali)-[~/Desktop]
$
```

Part B - Examine Embedded Metadata-Like Content

Use xxd and strings to inspect the JPEG for visible camera, software, name, date or other metadata-related strings.

Record only information actually found. If a searched item is not present, state **“not found”** rather than guessing.

Using xxd command to inspect the JPEG

```
(water@kali)~/Desktop
$ xxd J_ub_law.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*
00000010: 0000 0000 1000 0001 0100 0000 4013  ....@
00000020: 0000 0101 0100 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0100 0900 0000 e600 0000 1201  ....
00000060: 0100 0100 0000 0100 0000 1501 0100  ....
00000070: 0000 0300 0000 1a01 0100 0100 0000 ef00  ....
00000080: 0000 1b01 0100 0100 0000 f700 0000 2801  ....(
00000090: 0100 0100 0000 0200 0000 3101 0100 2100  ....1..!
000000a0: 0000 ff00 0000 3201 0100 1400 0000 2001  ....2....
000000b0: 0000 9882 0100 0c00 0000 3401 0000 9e9c  ....4....
000000c0: 0100 5200 0000 4001 0000 6987 0100 0100  ....R...@...i
000000d0: 0000 9401 0000 4404 0000 0800 0100 0800  ....D....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f  NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010  N.NIKON D4...-.
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265  '...-.'.Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120  Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230  (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500  :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000  HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000  1.3,...c.a.m.p.
00000160: 7500 7300 2c00 2000 4d00 6500 7200 7200  u.s.,,M.e.r.r.r.
00000170: 6900 6300 6b00 2000 5300 6300 6800 6f00  i.c.k..S.c.h.o.
00000180: 6f00 6c00 2000 6f00 6c00 2000 4200 7500  o.l..o.f..B.u.
00000190: 7300 6900 6e00 6500 7300 7300 0000 0000  s.i.n.e.s.s....
000001a0: 2900 9a82 0100 0100 0000 8603 0000 9d82  ).
000001b0: 0100 0100 0000 8e03 0000 2288 0100 0100  ....
```

Using String command to inspect the JPEG

```
(water@kali)~/Desktop
$ strings -a J_ub_law.jpg | grep -iE "camera|make|model|canon|nikon|sony|iphone|samsung"
NIKON CORPORATION
NIKON D4
" id="W5MOMpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38" > <rdf:RDF xmlns:rdf="
http://www.w3.org/1999/02-22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="http://purl.org/dc/elements/1.1/
" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns
.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns
:lr="http://ns.adobe.com/lightroom/1.0/" aux:ApproximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99" aux:Lens
Info="170/10 350/10 28/10 28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DateCreated="2013-10-08" photoshop:LegacyIPTCDigest="5E379AE71DB032222F
A2DE651E47F460" photoshop:ColorMode="3" photoshop:ICCPProfile="Adobe RGB (1998)" xmp:CreateDate="2013-10-08T18:56:21" xmp:CreatorTool="NIKON D4 Ver.1.01" xmp:Label=
"Approved" xmp:MetadataDate="2020-08-20T10:20:05-04:00" xmp:ModifyDate="2020-08-20T10:20:05-04:00" xmp:Rating="0" xmpMM:DocumentID="adobe:docid:photoshop:307cd9db-3
a87-6148-9874-c6730fd42952" xmpMM:InstanceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" xmpMM:OriginalDocumentID="B04B02E367C62737F8E78649729D6364" crs:AlreadyA
pplied="True" crs:AutoLateralCA="0" crs:AutoWhiteVersion="134348800" crs:Blacks2012="0" crs:BlueHue="0" crs:BlueSaturation="0" crs:CameraProfile="Embedded" crs:Cam
eProfileDigest="54650A34185B5CCA58442D0B43A92BCE" crs:Clarity2012="0" crs:ColorNoiseReduction="0" crs:Contrast2012="0" crs:ConvertToGrayscale="False" crs:Defringe6
```

```
(water@kali)~/Desktop
$ strings -a J_ub_law.jpg | grep -iE "software|adobe|photoshop|lightroom|gimp"
Adobe Photoshop 21.1 (Macintosh)
Adobe_CM
Adobe
"Photoshop 3.0
Adobe_CM
Adobe
```

```

(water@kali)-[~/Desktop]
└─$ strings -a J_ub_law.jpg | grep -iE "name|author|artist|creator"
printerName TEXT
" id="WSM0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38" ><rdf:RDF xmlns:rdf="
http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="http://purl.org/dc/elements/1.1/
" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns
.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns

```

```

(water@kali)-[~/Desktop]
└─$ strings -a J_ub_law.jpg | grep -iE "date|202[0-9]|201[0-9]|20[0-9][0-9]"
2020:08:20 10:20:05
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
20131008
2013
" id="WSM0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38" ><rdf:RDF xmlns:rdf="
http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="http://purl.org/dc/elements/1.1/
" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns
.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns

```

```

(water@kali)-[~/Desktop]
└─$ cat jpeg_strings.txt
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
HOWARD KORN
0221
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
17.0-35.0 mm f/2.8
Adobe_CM
Adobe
b34r

```

Part C - Direct Data-Unit Examination with The Sleuth Kit

Use tools such as:

img_stat

mmls

fsstat

fls

icat

blkcat

Identify actual inode and sector/block values from your own command output. Do not copy placeholder values from examples.

```

(water@kali)-[~/Desktop]
└─$ ls
120M-7z      File_carving.docx  jpeg_strings.txt  knock-env  'lab file'  reconstructed.jpg  usb
CIP-B101_Lab1C_wale_water  image_hex.txt      J_ub_law.jpg      lab         NTFS.zip    _tower_original_image_for_lab.bmp

(water@kali)-[~/Desktop]
└─$ cd usb

(water@kali)-[~/Desktop/usb]
└─$ ls
Ch01InChap01.dd

(water@kali)-[~/Desktop/usb]
└─$

```


Image Information

```
(water@kali)-[~/Desktop/usb]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512

(water@kali)-[~/Desktop/usb]
$
```

Part D - Embedded File Discovery with Binwalk

Use binwalk to identify embedded file structures or signatures.

If the instructor-provided File_carving.docx is available, perform the demonstrated manual and automatic extraction workflow. If it is not available, document this limitation and complete the Binwalk practice using the other supplied lab evidence.

```
(water@kali)-[~/Desktop]
$ ls
128M.7z      File_carving.docx  jpeg_strings.txt  knock-env  'lab file'  reconstructed.jpg  usb
CIP-B101_Lab1C_wale_water  image_hex.txt      J_ub_law.jpg      lab         NTFS.zip    _tower_original_image_for_lab.bmp

(water@kali)-[~/Desktop]
$ binwalk File_carving.docx

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0          Zip archive data, at least v2.0 to extract, compressed size: 358, uncompressed size: 1364, name: [Content_Types].xml
927          0x39F        Zip archive data, at least v2.0 to extract, compressed size: 239, uncompressed size: 590, name: _rels/.rels
1727         0x6BF        Zip archive data, at least v2.0 to extract, compressed size: 328, uncompressed size: 1149, name: word/_rels/document.xml.rels
2377         0x949        Zip archive data, at least v2.0 to extract, compressed size: 1458, uncompressed size: 4120, name: word/document.xml
3882         0xF2A        Zip archive data, at least v1.0 to extract, compressed size: 85046, uncompressed size: 85046, name: word/media/image1.jpeg
88980        0x15B94      Zip archive data, at least v2.0 to extract, compressed size: 1571, uncompressed size: 6795, name: word/theme/theme1.xml
90602        0x161EA      Zip archive data, at least v2.0 to extract, compressed size: 1092, uncompressed size: 2785, name: word/settings.xml
91741        0x1665D      Zip archive data, at least v2.0 to extract, compressed size: 265, uncompressed size: 497, name: word/webSettings.xml
92056        0x16798      Zip archive data, at least v2.0 to extract, compressed size: 369, uncompressed size: 749, name: docProps/core.xml
92736        0x16A40      Zip archive data, at least v2.0 to extract, compressed size: 2927, uncompressed size: 28961, name: word/styles.xml
95708        0x175DC      Zip archive data, at least v2.0 to extract, compressed size: 453, uncompressed size: 1261, name: word/fontTable.xml
96209        0x177D1      Zip archive data, at least v2.0 to extract, compressed size: 388, uncompressed size: 731, name: docProps/app.xml
97680        0x17D90      End of Zip archive, footer length: 22

(water@kali)-[~/Desktop]
$
```

```
(water@kali)-[~/Desktop]
$ dd if=File_carving.docx skip=3934 of=coded_again.jpg count=85046 bs=1
85046+0 records in
85046+0 records out
85046 bytes (85 kB, 83 KiB) copied, 0.342986 s, 248 kB/s

(water@kali)-[~/Desktop]
$ ls
128M.7z      coded_again.jpg  image_hex.txt  J_ub_law.jpg  lab  NTFS.zip  _tower_original_image_for_lab.bmp
CIP-B101_Lab1C_wale_water  File_carving.docx  jpeg_strings.txt  knock-env  'lab file'  reconstructed.jpg  usb

(water@kali)-[~/Desktop]
$
```



```

(water@kali)-[~/Desktop]
$ binwalk File_carving.docx

```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	Zip archive data, at least v2.0 to extract, compressed size: 358, uncompressed size: 1364, name: [Content_Types].xml
927	0x39F	Zip archive data, at least v2.0 to extract, compressed size: 239, uncompressed size: 590, name: _rels/.rels
1727	0x68F	Zip archive data, at least v2.0 to extract, compressed size: 328, uncompressed size: 1149, name: word/_rels/document.xml.rels
2377	0x949	Zip archive data, at least v2.0 to extract, compressed size: 1458, uncompressed size: 4120, name: word/document.xml
3882	0xF2A	Zip archive data, at least v1.0 to extract, compressed size: 85046, uncompressed size: 85046, name: word/media/image1.jpeg
88980	0x15B94	Zip archive data, at least v2.0 to extract, compressed size: 1571, uncompressed size: 6795, name: word/theme/theme1.xml
90602	0x161EA	Zip archive data, at least v2.0 to extract, compressed size: 1092, uncompressed size: 2785, name: word/settings.xml
91741	0x1665D	Zip archive data, at least v2.0 to extract, compressed size: 265, uncompressed size: 497, name: word/webSettings.xml
92056	0x16798	Zip archive data, at least v2.0 to extract, compressed size: 369, uncompressed size: 749, name: docProps/core.xml
92736	0x16A40	Zip archive data, at least v2.0 to extract, compressed size: 2927, uncompressed size: 28961, name: word/styles.xml
95708	0x175DC	Zip archive data, at least v2.0 to extract, compressed size: 53, uncompressed size: 1261, name: word/fontTable.xml
96209	0x177D1	Zip archive data, at least v2.0 to extract, compressed size: 88, uncompressed size: 731, name: docProps/app.xml
97680	0x17D90	End of Zip archive, footer length: 22

```

(water@kali)-[~/Desktop]
$ dd if=File_carving.docx skip=3934 of=coded_again.jpg count=85046 bs=1
85046+0 records in
85046+0 records out
85046 bytes (85 kB, 83 KiB) copied, 0.342986 s, 248 kB/s

(water@kali)-[~/Desktop]
$ ls
120M.7z          coded_again.jpg  image_hex.txt    J_ub_lab.jpg    lab             NTFS.zip         _tower_original_image_for_lab.bmp
CIP-B101_lab1C_wale_water  File_carving.docx  jpeg_strings.txt  knock-env       'lab file'      reconstructed.jpg  usb

(water@kali)-[~/Desktop]
$ display coded_again.jpg

```

Part E - Prepare the USB Image

Extract 120M.7z, identify the resulting USB image, calculate hashes, inspect the image using TSK, and confirm the appropriate partition offset using mmls.

The source lab refers to an offset of 128, but students must confirm the partition structure rather than blindly copying the value.

Checking the file info

```

120M.7z  cacert.der  xampp-linux-x64-8.2.4-0-installer.run

(water@kali)-[~/Downloads]
$ ls -l 120M.7z
-rw-rw-r-- 1 water water 36720470 Sep  4 08:22 120M.7z

```

Calculating the Hash

```

(water@kali)-[~/Downloads]
$ hashdeep -c md5,sha1 120M.7z
##### HASHDEEP-1.0
##### size,md5,sha1,filename
## Invoked from: /home/water/Downloads
## $ hashdeep -c md5,sha1 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2810745018afaa2da31dc17a8eb590fca66eeef7,/home/water/Downloads/120M.7z

```

Unzipping the File

```
(water@kali) [~/Downloads]
$ 7z l 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

  Date       Time       Attr      Size    Compressed  Name
-----
2021-09-22  22:59:31  D...A        0          0  120M
2021-09-22  22:59:31  ....A 124780544    36720256  120M/usb_fat_carving.001
2021-09-22  22:59:32  ....A    1685          1685  120M/usb_fat_carving.001.txt
-----
2021-09-22  22:59:32          124782229    36720256  2 files, 1 folders

(water@kali) [~/Downloads]
$
```

Extracting the content of the file

```
(water@kali) [~/Downloads]
$ 7z e 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470
```

Examine the content of the usb

```
(water@kali) [~/Downloads]
$ ls
120M 120M.7z  cacert.der  usb_fat_carving.001  usb_fat_carving.001.txt  xampp-linux-x64-8.2.4-0-installer.run
```

```
(water@kali) [~/Downloads]
$ find -o 128 usb_fat_carving.001
r/r 3: USB (Volume Label Entry)
d/d 6: System Volume Information
r/r * 7: _est
r/r * 10: .dropbox_device
d/d * 13: old_File_Carving_files
r/r * 15: B_ub_poe4.bmp
r/r * 18: B_zoom-eubie-mono.bmp
r/r * 21: Ballardlab8.java
r/r * 24: brittlab10.java
r/r * 27: DO_example.doc
r/r * 30: DO_example2.doc
r/r * 33: G_BuiltForThis.gif
r/r * 35: G_zoom-sc.gif
r/r * 37: H_Form.html
r/r * 39: H_hello.html
r/r * 41: J_ub_law.jpg
r/r * 44: J_ub_night.jpg
r/r * 47: nps-2008-jean_outlook.pst
r/r * 50: P_CAS-zoom-6.png
r/r * 53: P_MSB_1_zoom.png
r/r * 57: pd_evidence_search_techniques.pdf
r/r * 61: pd_Forensic_Report_Template.pdf
r/r * 64: pp_Number_Systems.pptx
r/r * 67: pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71: _tf_1.rtf
r/r * 74: T_Eubie-iphone-5-8.tiff
r/r * 78: T_youknowus-iphone-5-8.tiff
r/r * 81: W_axloadcomplete.wav
r/r * 84: W_speaker_test_sound.wav
r/r * 86: Z_file.zip
r/r * 88: 江莎莎行踪轨迹.doc
r/r * 91: nps-2008-jean_outlook.pst
v/v 3889667: $MBR
v/v 3889668: $FAT1
v/v 3889669: $FAT2
v/v 3889670: $OrphanFiles
```

```
(water@kali)-[~/Downloads]
```

[illegible]

```

(water@kali)-[~/Desktop]
$ scalpel usb_fat_carving.001 -o output
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/water/Desktop/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" --> 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" --> 9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 1 seconds.

(water@kali)-[~/Desktop]
$

```

```

(water@kali)-[~/Desktop]
$ cd output

(water@kali)-[~/Desktop/output]
$ ls
audit.txt  jpg-0-0  jpg-1-0

(water@kali)-[~/Desktop/output]
$

```

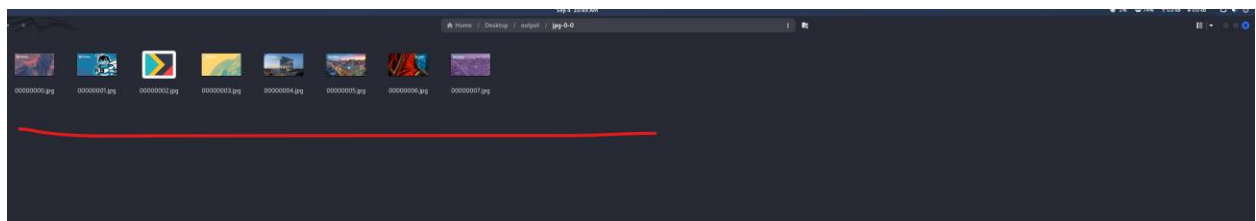
```

(water@kali)-[~/Desktop/output]
$ cd jpg-0-0

(water@kali)-[~/Desktop/output/jpg-0-0]
$ ls
00000000.jpg 00000001.jpg 00000002.jpg 00000003.jpg 00000004.jpg 00000005.jpg 00000006.jpg 00000007.jpg

(water@kali)-[~/Desktop/output/jpg-0-0]
$

```



```

(water@kali)-[~/Desktop]
$ tree output
output
├── audit.txt
├── jpg-0-0
│   ├── 00000000.jpg
│   ├── 00000001.jpg
│   ├── 00000002.jpg
│   ├── 00000003.jpg
│   ├── 00000004.jpg
│   ├── 00000005.jpg
│   ├── 00000006.jpg
│   └── 00000007.jpg
└── jpg-1-0
    ├── 00000008.jpg
    ├── 00000009.jpg
    ├── 00000010.jpg
    ├── 00000011.jpg
    ├── 00000012.jpg
    ├── 00000013.jpg
    ├── 00000014.jpg
    ├── 00000015.jpg
    └── 00000016.jpg

3 directories, 18 files

(water@kali)-[~/Desktop]
$

```

2.7. Analysis and Findings

The analysis showed how forensic tools can be used to examine raw data and recover files when normal file-system access is not reliable. XXD was used to examine hexadecimal data and identify JPEG file signatures, while The Sleuth Kit was used to inspect the forensic image at the inode, block, and sector levels. Binwalk was used to identify embedded file structures, and Scalpel was used to recover deleted files from the USB image. The recovered evidence was reviewed and hashes were used to help verify the integrity of the files.

2.8. Challenges and Solutions

During the practical, some commands produced errors, especially when working with extraction and Scalpel output directories. One issue was using a non-empty output directory for Scalpel, which prevented the carving process from running correctly. This was solved by using a new or empty output directory before running the command again. Another challenge was confirming the correct partition offset instead of relying on the example value, so mmls was used to examine the image and confirm the partition structure.

2.9. Conclusion

This lab improved my understanding of digital forensic data carving and file recovery. I learned how to examine binary data using XXD, identify file signatures, analyse forensic images with The Sleuth Kit, find embedded content with Binwalk, and recover deleted files using Scalpel. I also learned the importance of hashing evidence and documenting commands, results, errors, and limitations. Overall, the practical showed how forensic tools can help recover useful evidence from damaged or deleted files.

2.10. Recommendations

Regular Assignment like this will help students improve their knowledge

2.11. References

Class Video Guide from Founder ICDFA on Computer and Digital Forensic

Digital Forensics Basics: A Step-by-Step Guide for Beginners — Companion Materials guide from [GitHub - frankwxu/Digital-Forensics-Basic-Book](#) · [GitHub](#)